

GRAFT-VERSUS-HOST DISEASE

Intestinal goblet cells protect against GVHD after allogeneic stem cell transplantation via Lypd8

Takahide Ara¹, Daigo Hashimoto^{1*}, Eiko Hayase¹, Clara Noizat¹, Ryo Kikuchi¹, Yuta Hasegawa¹, Kana Matsuda², Shoko Ono², Yoshihiro Matsuno³, Ko Ebata¹, Reiki Ogasawara¹, Shuichiro Takahashi¹, Hiroyuki Ohigashi¹, Emi Yokoyama¹, Keitaro Matsuo^{4,5}, Junichi Sugita¹, Masahiro Onozawa¹, Ryu Okumura⁶, Kiyoshi Takeda⁶, Takanori Teshima^{1*}

Copyright © 2020
The Authors, some
rights reserved;
exclusive licensee
American Association
for the Advancement
of Science. No claim
to original U.S.
Government Works

Graft-versus-host disease (GVHD) and infection are major obstacles to successful allogeneic hematopoietic stem cell transplantation (HSCT). Intestinal goblet cells form the mucus layers, which spatially segregate gut microbiota from host tissues. Although it is well known that goblet cell loss is one of the histologic features of GVHD, effects of their loss in pathophysiology of GVHD remain to be elucidated. In mouse models of allogeneic HSCT, goblet cells in the colon were significantly reduced, resulting in disruption of the inner mucus layer of the colon and increased bacterial translocation into colonic mucosa. Pretransplant administration of interleukin-25 (IL-25), a growth factor for goblet cells, protected goblet cells against GVHD, prevented bacterial translocation, reduced plasma concentrations of interferon- γ (IFN- γ) and IL-6, and ameliorated GVHD. The protective role of IL-25 was dependent on Lypd8, an antimicrobial molecule produced by enterocytes in the colon that suppresses motility of flagellated bacteria. In clinical colon biopsies, low numbers of goblet cells were significantly associated with severe intestinal GVHD, increased transplant-related mortality, and poor survival after HSCT. Goblet cell loss is associated with poor transplant outcome, and administration of IL-25 represents an adjunct therapeutic strategy for GVHD by protecting goblet cells.

INTRODUCTION

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is a curative therapy for various hematologic malignant and nonmalignant disorders. However, widespread use of allo-HSCT is still limited by complications, such as graft-versus-host disease (GVHD) and infections, both of which are associated with considerable morbidity and mortality after allo-HSCT. Acute GVHD is an alloreactive donor T cell-mediated inflammatory disease after allo-HSCT, primarily involving the skin, liver, and gastrointestinal (GI) tract (1). The major targets of GVHD in the GI tract are mature and progenitors of mucosal epithelial cells, including intestinal stem cells (2).

Trillions of microbes inhabit humans, with the largest population in the GI tract, particularly in the colon (3). Emerging evidence suggests an association of intestinal dysbiosis and GVHD after allo-HSCT (4–6). There are two types of intestinal secretory cells that play a protective role against pathogenic bacteria in the gut lumen: Paneth cells in the small intestine and goblet cells that mainly reside in the colon. Paneth cells are essential regulators of the composition of intestinal microbiota by secreting various antimicrobial peptides (AMPs) such as α -defensins (7–9). We previously reported that GVHD-mediated Paneth cell injury results in substantially reduced production of α -defensins in the small intestine and perturbed intestinal microbiota in mice (6, 10). In humans, Paneth cell count

in duodenal biopsy negatively correlates with severity and outcome of GI-GVHD (11).

The majority of bacteria of the human inhabit in the colon; bacterial content reaches as high as 10^{12} cells per gram of feces in the colon, emphasizing the importance of mucosal barrier function in the large intestine against bacterial translocation (12). Goblet cells secrete mucus that shields the gut epithelium from bacteria. While the small intestine has only single mucus layer, the large intestine has a two-mucus-layered system, which serves as a chemical and mechanical barrier against pathogens. The inner mucus layer is firmly attached to the epithelial cells with high AMP concentration and spatially separates the commensal bacteria from epithelium, while the outer mucus layer provides a habitat for commensal microbiota (13–15). Although decrease in number of goblet cells is one of the pathological features of GI-GVHD (16, 17), dynamic process of goblet cell injury after allo-HSCT and a role of goblet cells in pathophysiology of GI-GVHD have not been well understood. In the current study, we explored the pathophysiological role of intestinal goblet cells in GVHD using mouse HSCT models and human colon biopsy samples. Furthermore, we explored the therapeutic potential of expanding intestinal goblet cells in an experimental GVHD model.

RESULTS

GVHD targets intestinal goblet cells, disrupts colonic mucus two-layered structure, and induces bacterial translocation after mouse allogeneic bone marrow transplantation

We first evaluated the fate of goblet cells after murine allogeneic bone marrow transplantation (BMT). B6D2F1 (H-2^{b/d}) mice were lethally irradiated with 12.5-Gy total body irradiation (TBI) and intravenously injected with 5×10^6 BM cells plus 5×10^6 splenocytes from major histocompatibility complex (MHC)-haploidentical allogeneic B6 (H-2^b) or syngeneic donors on day 0. Severe GVHD

¹Department of Hematology, Hokkaido University Faculty of Medicine and Graduate School of Medicine, Sapporo 060-8638, Japan. ²Department of Gastroenterology and Hepatology, Hokkaido University Graduate School of Medicine, Sapporo 060-8638, Japan. ³Department of Surgical Pathology, Hokkaido University Hospital, Sapporo 060-8648, Japan. ⁴Division of Cancer Epidemiology and Prevention, Aichi Cancer Center Research Institute, Nagoya 464-8681, Japan. ⁵Department of Cancer Epidemiology, Nagoya University Graduate School of Medicine, Nagoya 466-8550, Japan. ⁶Department of Microbiology and Immunology, Graduate School of Medicine, WPI Immunology Frontier Research Center, Osaka University, Suita 565-0871, Japan.
*Corresponding author. Email: teshima@med.hokudai.ac.jp (T.T.); D5hash@pop.med.hokudai.ac.jp (D.H.)

developed in allogeneic mice and 60% of allogeneic mice died by 7 weeks after SCT, whereas all syngeneic mice survived during this period ($P = 0.0010$; Fig. 1A). In allogeneic mice, clinical GVHD scores were significantly higher than those in syngeneic mice ($P < 0.005$; Fig. 1B). Alcian blue staining of the colon demonstrated that goblet cells were markedly reduced in allogeneic mice compared to syngeneic controls 5 days after BMT and thereafter ($P < 0.05$; Fig. 1, C and D). The proportion of matured goblet cells, defined as those with diameter greater than $10\ \mu\text{m}$ (18), was significantly reduced after allo-

geneic mice at 7 and 35 days after BMT ($P = 0.0022$ and $P = 0.0012$, respectively), suggesting that GVHD preferentially targets mature goblet cells or inhibits maturation of goblet cells (Fig. 1E). To exclude the possible role of TBI in goblet cell loss after BMT, we also assessed the fate of goblet cells after chemotherapy-conditioned BMT. B6D2F1 mice were intraperitoneally injected with busulfan (25 mg/kg) from days -7 to -4 and cyclophosphamide (100 mg/kg) on days -3 and -2 , followed by intravenous injection with 5×10^6 BM cells plus 10×10^6 splenocytes from allogeneic B6 or syngeneic donors on day 0.

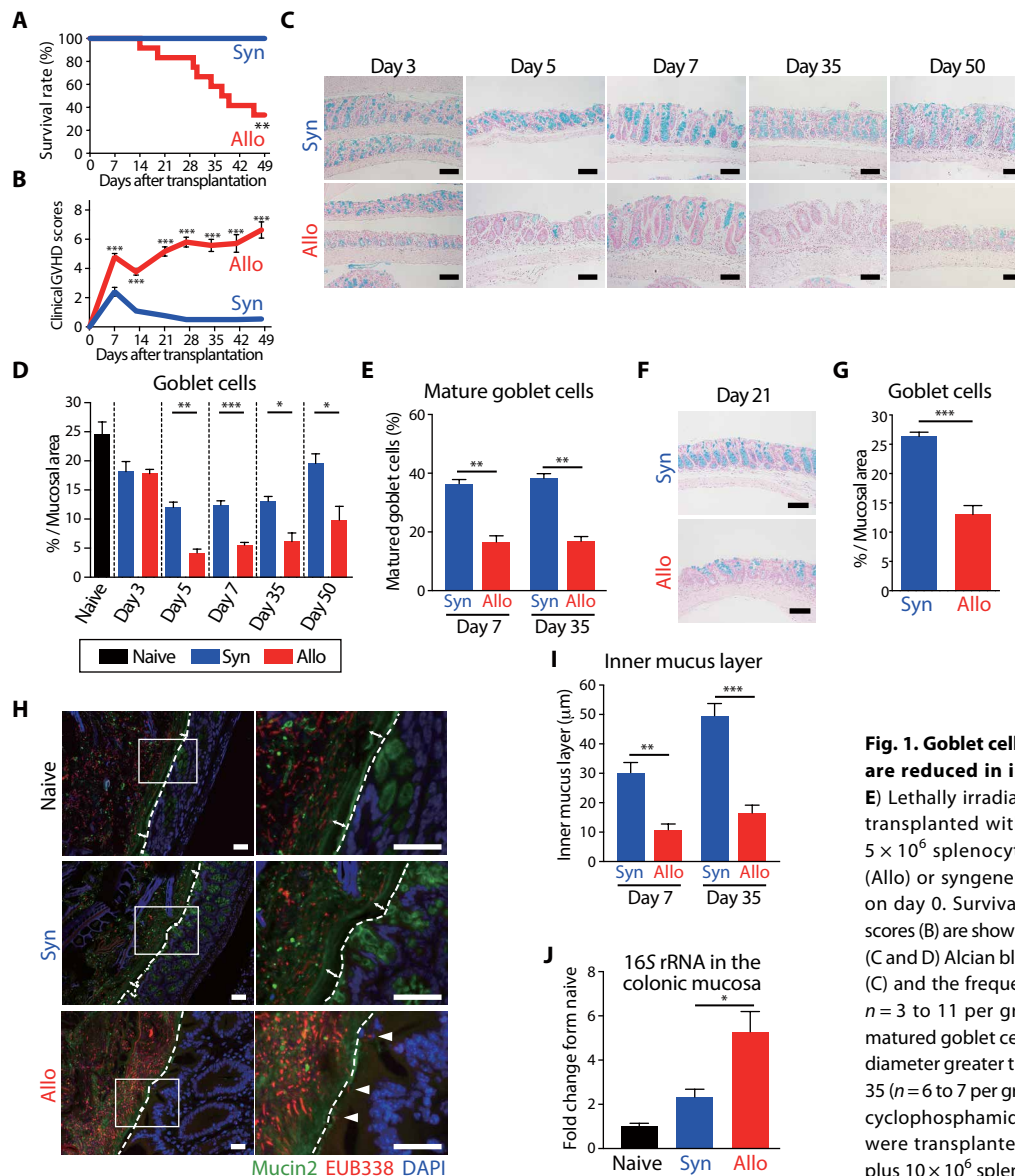


Fig. 1. Goblet cells in the large intestine are reduced in intestinal GVHD. (A to E) Lethally irradiated B6D2F1 mice were transplanted with 5×10^6 BM cells plus 5×10^6 splenocytes from allogeneic B6 (Allo) or syngeneic B6D2F1 (Syn) donors on day 0. Survival (A) and clinical GVHD scores (B) are shown ($n = 11$ to 12 per group). (C and D) Alcian blue staining of the colons (C) and the frequency of goblet cells [(D), $n = 3$ to 11 per group]. (E) Proportion of matured goblet cells defined as those with diameter greater than $10\ \mu\text{m}$ on days 7 and 35 ($n = 6$ to 7 per group). (F and G) Busulfan/cyclophosphamide-treated B6D2F1 mice were transplanted with 5×10^6 BM cells plus 10×10^6 splenocytes from B6 (Allo) or B6D2F1 (Syn) donors. Alcian blue staining

of the colons (F) and the frequency of goblet cells in the colon (G) on day +21 ($n = 9$ to 10 per group). (H) BMT was performed as in (A) to (E). Immunofluorescent staining of mucin-2 (green) with bacteria-targeting FISH (red) and 4',6-diamidino-2-phenylindole (DAPI) nuclear staining (blue) of the colon on day 35. Area in the white squares is magnified and shown in the right side. Double-headed arrows and arrowheads indicate the inner mucus layer and the bacteria located in the colonic mucosa, respectively. Dashed lines indicate the border between the inner mucus layer and epithelial cells. (I) Mean thickness of inner mucus layer on days 7 and 35 ($n = 6$ to 11 per group). (J) Bacterial DNA in the colonic lamina propria on day 7 ($n = 3$ to 10 per group). (A) Data from two independent experiments were combined, and log-rank test was used to compare survival curves. (B, D, E, G, I, and J) Data from two independent experiments were combined. Mann-Whitney U tests were used to compare the data. Scale bars, $100\ \mu\text{m}$. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.005$.

Again, goblet cells were significantly reduced on day 21 after chemotherapy-conditioned BMT ($P < 0.0001$; Fig. 1, F and G).

Goblet cells secrete mucus and construct mucus two-layered structure in the colon, which plays an important role to protect the colonic epithelial cells from pathogens. We investigated if goblet cell loss in GVHD could affect the mucous system. We performed immunofluorescent analysis of mucin-2 in combination with bacterial fluorescence in situ hybridization (FISH) to assess the integrity of mucus layers and bacterial translocation into the colonic mucosa 35 days after BMT. The inner mucus layer was significantly thinner in allogeneic mice than in syngeneic mice, and bacterial signals were observed in the lamina propria of the large intestine in allogeneic mice at 7 and 35 days after BMT ($P = 0.0047$ and $P < 0.0001$, respectively; Fig. 1, H and I). To quantify bacterial translocation into the colonic mucosa, DNA extracted from the lamina propria of the large intestine was subjected to quantitative polymerase chain reaction (qPCR) targeting 16S ribosomal RNA (rRNA). Bacterial load in the lamina propria was significantly increased in allogeneic mice compared to syngeneic controls ($P = 0.0133$; Fig. 1J). These data suggested that goblet cell loss in GVHD is associated with destruction of inner mucus layer and bacterial translocation to the intestinal tissues.

Interleukin-25 expands goblet cells in the large intestine and ameliorates GVHD

It has been reported that interleukin-25 (IL-25) supports the growth of goblet cells via a type 2 innate lymphoid cell (ILC2)-dependent manner (19, 20). We found that plasma concentrations of IL-25 measured using enzyme-linked immunosorbent assay (ELISA) were not increased up to 72 hours after TBI but significantly elevated on day 7 after SCT without a difference between syngeneic and allogeneic SCT (fig. S1, A and B). These results suggest that goblet cell loss after allogeneic SCT was not attributed to the insufficiency of IL-25. We confirmed that administration of IL-25 for 7 days significantly increased goblet cells in the large intestine of B6D2F1 mice ($P = 0.0262$; Fig. 2, A and B). Periodic acid Schiff staining showed that goblet cells with mature morphology were substantially increased, suggesting that IL-25 promotes not only the differentiation of goblet cells but also maturation of them ($P < 0.0001$; Fig. 2C). When IL-25 was administered from days -6 to 0 of allogeneic BMT, significantly greater number of goblet cells persisted on day 5 after BMT compared to

allogeneic controls ($P = 0.0002$; Fig. 2, D and E). qPCR for 16S rRNA of the colon demonstrated that bacterial load in the lamina propria was significantly increased in allogeneic mice compared to syngeneic controls, whereas IL-25 significantly reduced bacterial translocation in allogeneic mice ($P = 0.0205$) and bacterial load in IL-25-treated allogeneic mice was comparable to the amount in syngeneic mice (Fig. 2F). Plasma concentrations of the proinflammatory cytokines interferon- γ (IFN- γ) and IL-6 were considerably increased in allogeneic mice compared to syngeneic controls, whereas IL-25 significantly suppressed IFN- γ and IL-6 production in allogeneic recipients ($P = 0.0056$ and $P = 0.0273$, respectively; Fig. 2, G and H). In sharp

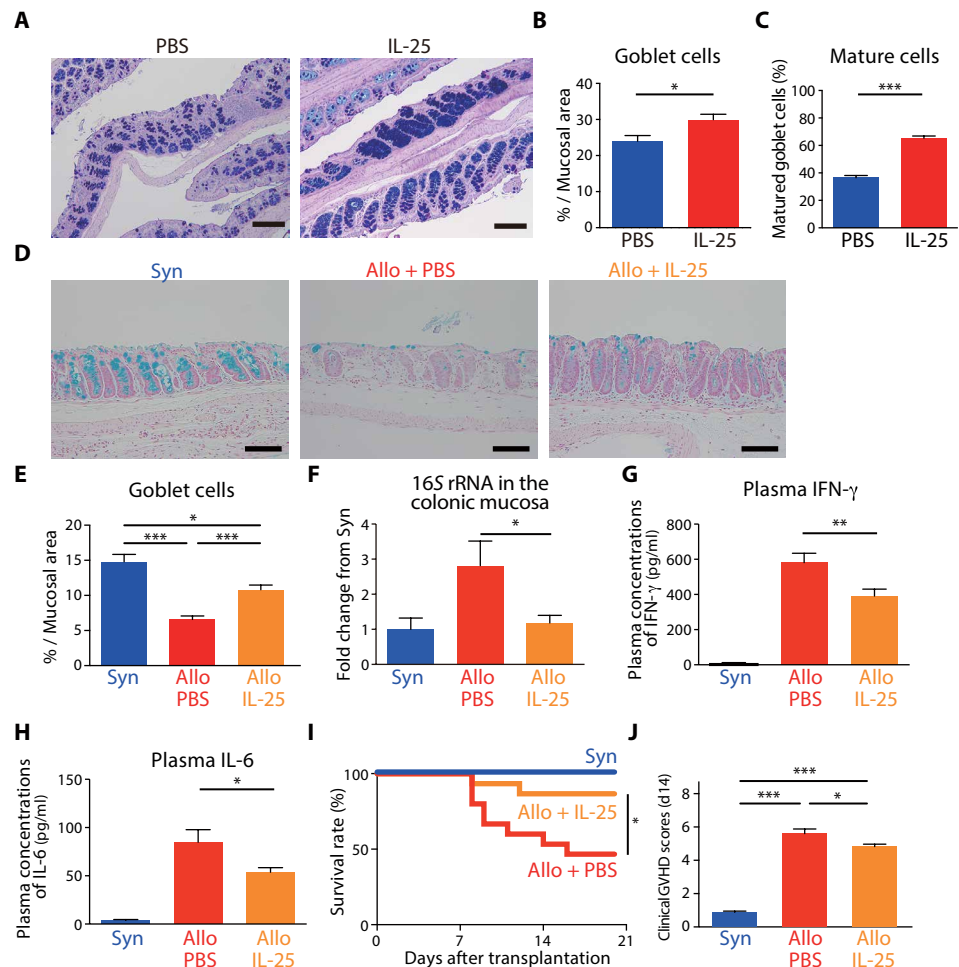


Fig. 2. IL-25 promotes the proliferation of goblet cells in the large intestine and attenuates GVH reaction. (A to C) Naïve B6D2F1 mice were intravenously injected with phosphate-buffered saline (PBS) or rmlIL-25 for 7 days, and the colons were harvested a day after the last injection. Periodic acid Schiff staining of the colons (A), frequency of goblet cells (B) ($n = 7$ per group), and the proportion of mature goblet cells among total goblet cells (C) are shown ($n = 8$ per group). (D to J) Lethally irradiated B6D2F1 mice were transplanted with 5×10^6 BM cells plus 10×10^6 splenocytes from allogeneic B6 donors on day 0. A group of allogeneic recipients was intravenously injected with rmlIL-25 from days -6 to 0 before HSCT. Alcian blue staining (D) and frequency of goblet cells in the large intestines (E) on day 5 ($n = 6$ to 9 per group). (F) qPCR targeting bacterial 16S rRNA gene was performed using total DNA extracted from the epithelial cell-removed large intestine on day 5. Fold differences relative to the 16S rRNA quantities of syngeneic controls ($n = 6$ to 8 per group). Plasma concentrations of IFN- γ (G) and IL-6 (H) from four independent experiments were combined ($n = 12$ to 19 per group). Survival probabilities (I) and clinical GVHD scores on day 14 (J) were shown ($n = 6$ to 15 per group). (B, C, E, F, and J) Data from two independent experiments were combined. Mann-Whitney U tests were used to compare the data. (I) Log-rank test was used to compare survival curves. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.005$.

contrast, IL-25 did not increase goblet cells persisting on day 5 or decrease bacterial translocation or proinflammatory cytokines after syngeneic SCT (fig. S2, A to E). These results suggest that IL-25 protects goblet cells against GVHD. In association with reduced bacterial translocation and proinflammatory cytokines, survival rates and GVHD scores were moderately ameliorated in IL-25-treated allogeneic recipients compared to allogeneic controls ($P = 0.0227$ and $P = 0.0133$, respectively; Fig. 2, I and J). We also evaluated the effects of IL-25 in another HSCT model, where the B6 mice were transplanted from BALB/c donors. Again, IL-25 significantly mitigated morbidity and mortality of GVHD ($P = 0.0059$), in association with less severe body weight loss, ruling out strain-dependent effects of IL-25 (fig. S2, F to H). Together, these data indicate that short-term administration of IL-25 protects goblet cells and ameliorated inflammatory responses in GVHD.

Lypd8 in the mucus layer is reduced in GVHD

We and others previously found that *Escherichia* was expanded in the gut after mouse allo-HSCT (6, 21) and intestinal dysbiosis accompanied by *Enterobacteriaceae* expansion correlated with mortality after clinical transplantation (22). Recently, it has been shown that Lypd8 produced by mucin-2-negative colon epithelial cells plays a critical role in segregation of flagellated bacteria such as *Escherichia coli* from colonic epithelia (14). Immunofluorescence analysis demonstrated that Lypd8 protein was enriched at the border between the inner mucus layer and colonic epithelial cells in naïve and syngeneic animals but reduced in allogeneic recipients (Fig. 3A). qPCR showed that mRNA expression of *Lypd8* in the colonic epithelial cells was significantly higher than in the lamina propria and serosal layer ($P = 0.0286$) and was comparable between syngeneic and allogeneic animals, indicating that *Lypd8* expression on a per-cell basis was not altered in GVHD (Fig. 3B).

Lypd8 in the mucus layer inhibits bacterial translocation and attenuates GVHD

On the basis these findings, we next tested if Lypd8 plays a protective role against bacterial translocation after allo-HSCT by using

Lypd8-deficient (*Lypd8*^{-/-}) mice as recipients. We found that bacterial load in the epithelium-removed colonic mucosa was comparable between naïve wild-type (WT) and B6-*Lypd8*^{-/-} mice, suggesting that Lypd8 is dispensable to prevent bacterial translocation into colonic mucosa in naïve mice with healthy microflora (Fig. 4A). Lethally irradiated B6-*Lypd8*^{-/-} mice and B6-WT mice were transplanted with BM cells and splenocytes from BALB/c mice. Bacterial FISH demonstrated more bacterial signals in the colonic lamina propria of *Lypd8*^{-/-} recipients compared to that of WT recipients after allogeneic BMT (Fig. 4B). Bacterial load determined by abundance of 16S rRNA in colonic mucosa was significantly greater in *Lypd8*^{-/-} recipients compared to WT recipients ($P = 0.0317$; Fig. 4C), indicating that Lypd8 played a protective role against bacterial translocation into the colonic mucosa. Enhanced bacterial translocation in *Lypd8*^{-/-} recipients was associated with increased plasma concentrations of IFN- γ and IL-6 after allogeneic BMT ($P < 0.0001$ and $P < 0.0214$, respectively; Fig. 4, D and E). In this model, GVHD was also severe and 60% of WT recipients died by day 70. GVHD was significantly more severe in B6-*Lypd8*^{-/-} mice compared to B6-WT recipients, and all B6-*Lypd8*^{-/-} recipients died by day 70 ($P = 0.0026$; Fig. 4F). When B6-*Lypd8*^{-/-} and B6-WT recipients were cohoused for 4 weeks before BMT, GVHD was again significantly more severe in *Lypd8*^{-/-} mice compared to WT controls ($P = 0.0007$), indicating that different microbiota in B6-*Lypd8*^{-/-} and B6-WT mice were not responsible for the disparate severity of GVHD (Fig. 4G). Next, we tested if IL-25 treatment could alter production of Lypd8 in WT recipients. qPCR of the colonic epithelial cells showed that *Lypd8* expression was not altered by IL-25 treatment, whereas immunofluorescence staining demonstrated persistence of a Lypd8-enriched inner mucus layer in IL-25-treated allogeneic mice, suggesting that IL-25 protects the Lypd8-rich area by protecting the inner mucus layer (Fig. 4, H and I). IL-25 did not ameliorate GVHD in *Lypd8*^{-/-} recipients (Fig. 4J), suggesting that goblet cells play a protective role against GVHD via a Lypd8-dependent manner. When splenocytes in the graft were reduced to induce minimal mortality in *Lypd8*^{-/-} recipients, IL-25 again failed to suppress clinical GVHD scores or mitigate body weight loss, further confirming the critical role of Lypd8 in anti-

GVHD effects of IL-25 (fig. S2, I to K). Furthermore, IL-25 administered from days 1 to 6 after HSCT was not effective against GVHD (Fig. 4K). This was not surprising because IL-25 requires ILC2 to expand goblet cells and ILC2 is profoundly depleted after conditioning regimens (23). Together, we found that mouse GVHD targets goblet cells in the colon, whereas these cells play a protective role against bacterial translocation and GVHD via a Lypd8-dependent manner.

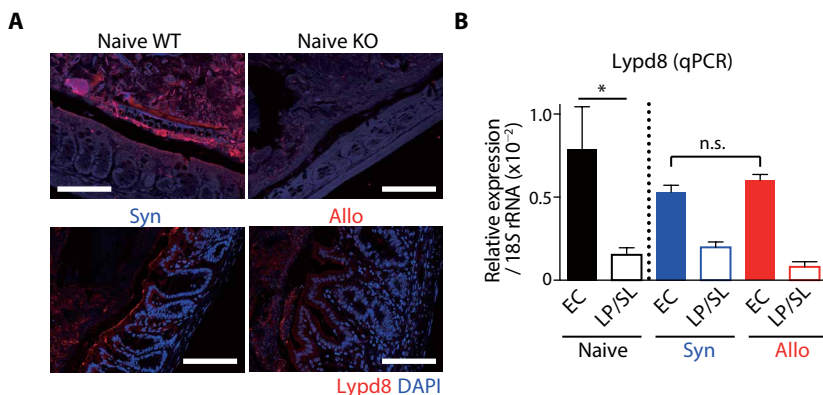


Fig. 3. Lypd8 in the mucus layer is reduced in GVHD. (A and B) B6 mice were lethally irradiated and transplanted with 5×10^6 BM cells plus 10×10^6 splenocytes from allogeneic BALB/c or syngeneic B6. (A) Lypd8 (red) with DAPI (blue) on day +7 after BMT or naïve wild-type (WT) and knockout (KO) mice. (B) Relative expression of Lypd8 to 18S rRNA in the epithelial cells (EC) or epithelial cell-removed colonic mucosa including lamina propria and serosal layer (LP/SL) ($n = 4$ to 10 per group). Data from two independent experiments were combined. Mann-Whitney U tests were used to compare the data. Scale bars, 100 μ m; N.S., not significant. * $P < 0.05$.

Colonic biopsy specimens from patients who underwent allo-HSCT

We then evaluated the fate of goblet cells after human allo-HSCT and the impact of goblet cell loss on the outcome of clinical transplantation. In 205 patients who underwent their first allo-HSCT in Hokkaido University Hospital between 2009 and 2015, incidence of GVHD was evaluable in 200 patients who had been successfully engrafted. Eighty-eight patients (42.9%) had 156 episodes requiring large intestine biopsies in the course of allo-HSCT. Table 1 shows characteristics of HSCT recipients with or without colon biopsy after

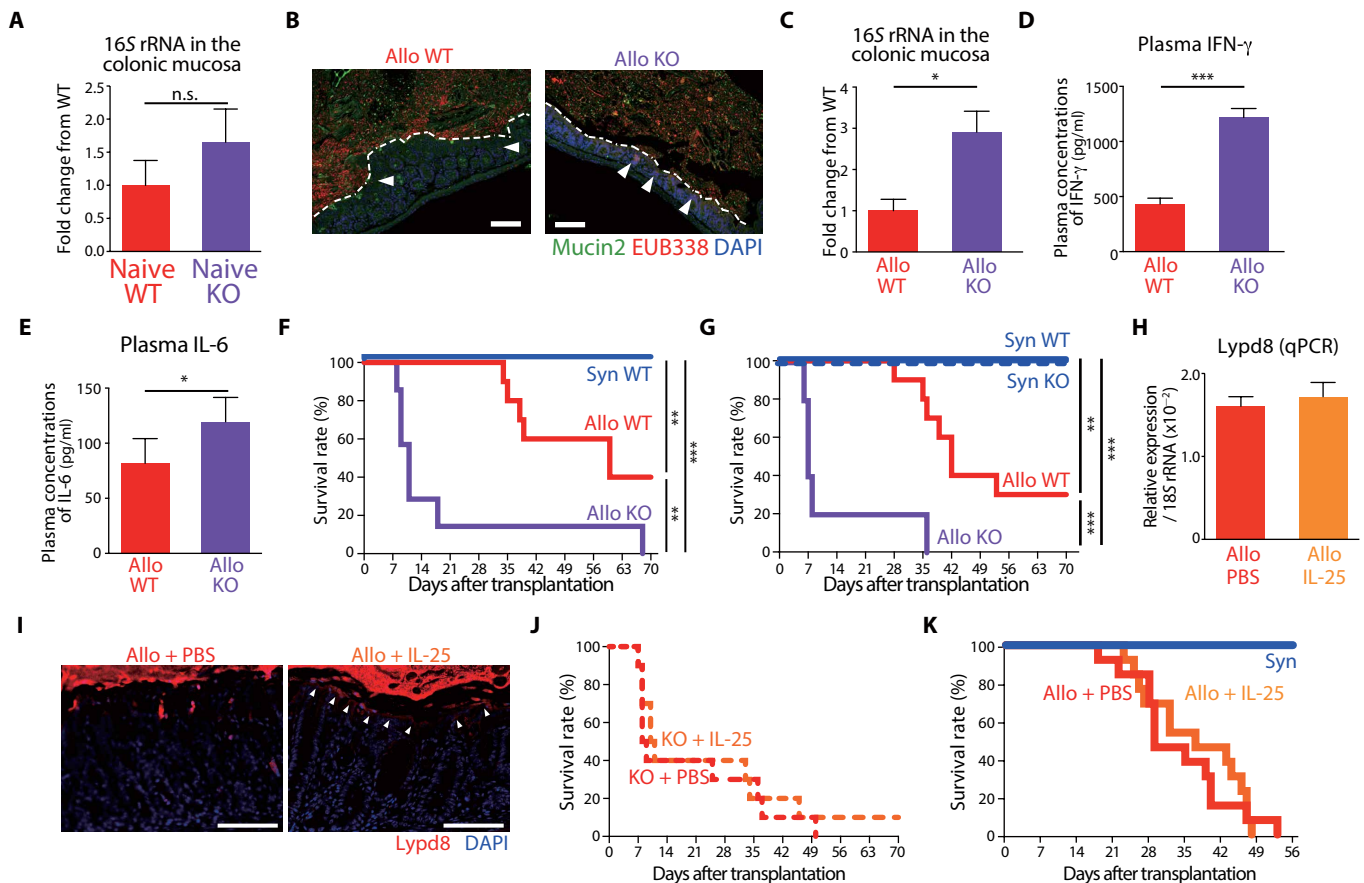


Fig. 4. Goblet cells suppress bacterial translocation via a Lypd8-dependent manner. (A) Bacterial 16S rRNA in the epithelial cell–removed colonic mucosa in naïve WT and *Lypd8*-deficient (KO) B6 mice. Fold differences relative to WT ($n = 7$ to 8 per groups). (B to F) WT or KO B6 mice were lethally irradiated and transplanted with 5×10^6 BM cells plus 10×10^6 splenocytes from allogeneic BALB/c or syngeneic B6. (B) Mucin-2 (green) with bacteria (red) and DAPI (blue) of the colon on day 7. Arrowheads indicate bacteria located in the colonic mucosa. Dashed lines indicate the border between the mucosal epithelium and gut lumen. (C) Bacterial 16S rRNA in the epithelial cell–removed colonic mucosa on day 7. Fold differences relative to WT ($n = 4$ to 5 per groups). Plasma concentrations of IFN- γ (D) and IL-6 (E) on day 5 ($n = 10$ to 13 per group). Survival curves (F) of WT and KO recipients ($n = 7$ to 10 per group). (G) Same setup as (B) to (F), but WT and KO recipients were cohoused from 4 weeks before HSCT. Survival curves ($n = 5$ to 10 per group). (H to J) Allogeneic WT (H and I) or KO (J) recipients ($n = 8$ to 10 per group) were intravenously injected with rIL-25 or PBS from days –6 to 0 before HSCT. Relative expression of *Lypd8* in colonic epithelial cells (H) and immunofluorescence analysis of *Lypd8* on day 5 (I) and survival curves (J). Arrowheads indicate the *Lypd8*-rich layer. (K) WT recipients were intravenously injected with rIL-25 or PBS from days 1 to 6 after HSCT. Survival curves ($n = 7$ to 13 per group). Data from two (A, C, and H) or three (D and E) independent experiments were combined. Mann-Whitney *U* tests were used to compare the data. Data from two (F and G) or three (J and K) independent experiments were combined, and log-rank tests were used to compare survival curves. Scale bars, 100 μ m; N.S., not significant. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.005$.

transplantation. Age, sex, primary diseases, donor sources, and the proportion of human leukocyte antigen (HLA)-matched transplantation of the patients who underwent colon biopsy were comparable to those of the total transplanted population, whereas significantly more patients with severe GVHD underwent colon biopsies. Biopsy specimens were categorized as summarized in fig. S3. Among 27 colon biopsies performed before transplantation, 22 biopsies were performed without any symptoms and categorized as the pre-HSCT screening group. Among 129 biopsies performed after allo-HSCT, 7 biopsies were excluded from further analyses due to the direct infiltration of adult T cell leukemia/lymphoma or posttransplantation lymphoproliferative disease, and the remaining 122 biopsies were analyzed. Among them, 109 biopsies were performed as initial diagnostic procedures for diarrheal episodes and categorized as initial biopsies, 7 biopsies performed due to per-

sistent diarrhea led to the different pathological diagnosis from the diagnosis at the initial biopsies and were considered as independent events from the previous events, and 6 biopsies were performed after successful treatment for pathologically confirmed GI-GVHD and categorized as follow-up biopsies. The initial biopsies consisted of 38 biopsies of pathologically confirmed GI-GVHD without cytomegalovirus (CMV) colitis (33 patients), 6 biopsies of pathologically confirmed GI-GVHD with CMV colitis (6 patients), 14 biopsies of clinical GI-GVHD without pathological finding of GVHD (12 patients), 9 biopsies of CMV colitis without GI-GVHD (8 patients), and 42 biopsies of nonspecific colitis (34 patients). The repeated biopsies performed due to persistent diarrhea consisted of a pathologically confirmed GI-GVHD, a pathologically confirmed GI-GVHD with CMV colitis, two episodes of CMV colitis (two patients), and three episodes of nonspecific colitis (two patients). All six follow-up

Table 1. Patient characteristics. AML, acute myeloid leukemia; ALL, acute lymphocytic leukemia; ML, malignant lymphoma; MDS, myelodysplastic syndrome; CML, chronic myelogenous leukemia; AA, aplastic anemia; MM, multiple myeloma; BMT, bone marrow transplantation; PBSCT, peripheral blood stem cell transplantation; CBT, cord blood transplantation; MAC, myeloablative conditioning; RIC, reduced-intensity conditioning.

Variable	Total population	Post-HSCT biopsy (–) population	Post-HSCT biopsy (+) population	P
	n = 205	n = 130	n = 75	
Age (median), years	17–67 (48)	17–67 (48)	18–67 (48)	0.987
Sex, n (%)				0.711
Male	117 (57)	73 (56)	44 (59)	
Female	88 (43)	57 (44)	31 (41)	
Underlying disease, n (%)				0.711
AML	76 (37)	58 (45)	18 (24)	
ALL	37 (18)	23 (18)	14 (19)	
ML	48 (23)	24 (18)	24 (32)	
MDS	23 (11)	13 (10)	10 (13)	
CML	6 (3)	4 (3)	2 (3)	
AA	6 (3)	2 (2)	4 (5)	
MM	2 (1)	0 (0)	2 (3)	
Others	7 (3)	6 (5)	1 (1)	
Disease risk index, n (%)				0.110
Low	25 (12)	13 (10)	12 (16)	
Intermediate	114 (56)	80 (62)	34 (45)	
High	53 (26)	31 (24)	22 (29)	
Very high	13 (6)	6 (5)	7 (9)	
Donor source, n (%)				0.278
BMT	112 (55)	66 (51)	46 (61)	
PBSCT	53 (26)	38 (29)	15 (20)	
CBT	40 (20)	26 (20)	14 (19)	
HLA match, n (%)				1
Yes	102 (50)	65 (50)	37 (49)	
No	103 (50)	65 (50)	38 (51)	
Conditioning, n (%)				0.057
MAC	117 (57)	81 (62)	36 (48)	
RIC	88 (43)	49 (38)	39 (52)	
TBI, n (%)				0.212
Yes	187 (91)	116 (89)	71 (95)	
No	18 (9)	14 (11)	4 (5)	
GVHD grade, n (%)	n = 200	n = 125	n = 75	<0.001
0–2	175 (88)	122 (98)	53 (71)	
3–4	25 (13)	3 (2)	22 (29)	
GI-GVHD stage, n (%)	n = 200	n = 125	n = 75	<0.001
0	147 (74)	122 (98)	25 (33)	
1–2	40 (20)	3 (2)	37 (49)	
3–4	13 (7)	0 (0)	13 (17)	

biopsies after GVHD treatment resulted in diagnosis of nonspecific colitis (six patients). There was no patient with pathologically confirmed upper GI-GVHD only.

Goblet cell damage in the large intestine correlates with inferior survival after clinical allo-HSCT

Histological examination demonstrated that a proportion of the numbers of goblet cells to total epithelial cells in the crypt of the large intestine were decreased in pathologically confirmed GI-GVHD cases ($20.0 \pm 1.8\%$) and pathologically confirmed GI-GVHD with CMV colitis cases ($17.0 \pm 3.0\%$) as compared to any other groups, including clinical GVHD patients (Fig. 5, A and B). The goblet cells in nonspecific colitis, clinical GI-GVHD, and CMV colitis alone after HSCT and pre-HSCT screening cases were not significantly different ($36.0 \pm 1.2\%$, $37.5 \pm 2.0\%$, $30.3 \pm 5.0\%$, and $37.2 \pm 1.2\%$, respectively). When only the first biopsy specimens after HSCT in each patient were analyzed, goblet cells were again significantly reduced in patients with pathologically confirmed intestinal GVHD (fig. S4). Furthermore, we found a statistical correlation between the number of goblet cells and the clinical severity of GI-GVHD ($P < 0.0001$; Fig. 5C). Next, we evaluated if corticosteroids could affect goblet cell loss and found that corticosteroid treatment at the time of biopsy

sampling did not affect goblet cell numbers; goblet cells were significantly reduced in pathologically confirmed GI-GVHD, regardless of steroid administration ($P < 0.0001$; Fig. 5D). Follow-up samples demonstrated that GVHD treatment significantly restored the goblet cells in the large intestine ($P = 0.0312$; Fig. 5E), suggesting that goblet cell numbers could potentially be a marker for monitoring during GVHD therapy.

Next, we evaluated the correlation between transplant outcomes and goblet cell reduction in large intestine. We defined severe goblet reduction as the percentage of goblet cells per crypt of 15% or less. We identified 8 patients who had severe goblet loss and 65 patients who had mild or no goblet loss. Patients' characteristics were comparable between the groups, except incidence of grade 3 to 4 GVHD that was significantly more prevalent in patients with severe goblet cell loss (Table 2). Overall survivals were significantly inferior in patients who underwent biopsies compared to those in patients who did not require colon biopsy ($P = 0.00794$ and $P < 0.0001$, respectively; Fig. 5F). Patients with severe goblet cell loss showed significantly shorter survival than those with mild or no goblet cell loss ($P = 0.0162$; Fig. 5F). Nonrelapse mortalities were significantly more in patients with severe goblet cell loss compared to those with mild or no goblet cell loss or patients without biopsy ($P = 0.0285$ and

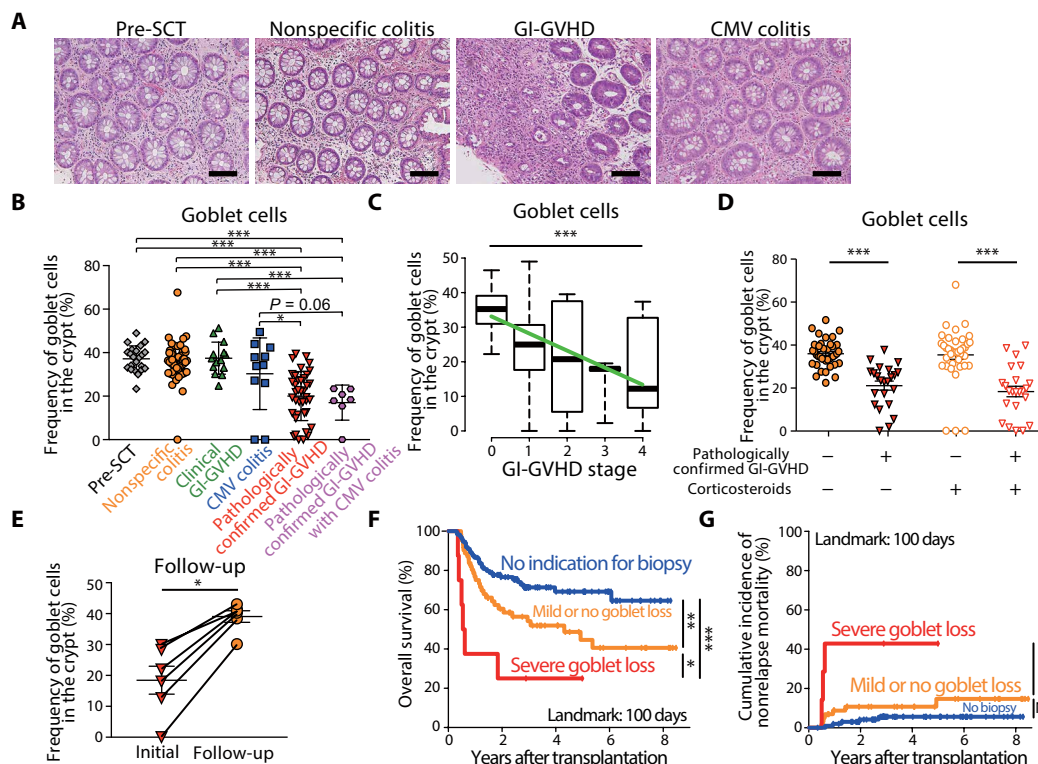


Fig. 5. Goblet cells in the large intestine decrease in human intestinal GVHD. Intestinal goblet cells were enumerated using colon biopsy specimens stained with H&E. (A) Representative H&E images of colon biopsy for pre-HSCT screening (pre-HSCT), or from recipients without GVHD (nonspecific colitis) on day 22, with pathologically confirmed GI-GVHD on day 29, or with CMV colitis on day 56 after HSCT. Magnification, $\times 200$. Scale bars, 100 μm . (B) Frequency of goblet cells within all nucleated cells in the crypt. (C) Proportion of goblet cells in patients without GI-GVHD (stage 0, $n = 23$) and stage 1 ($n = 32$), stage 2 ($n = 5$), stage 3 ($n = 5$), and stage 4 ($n = 8$) GI-GVHD. (D) Proportion of goblet cells in patients with or without corticosteroid treatment. (E) Proportion of goblet cells at follow-up biopsies after GVHD treatment and corresponding initial biopsies. (F and G) Landmark analysis at 100 days after HSCT was performed. The overall survival (F) and cumulative incidence of nonrelapse mortalities (G) of the patients with low proportion of goblet cells ($n = 8, 7$) and high proportion of goblet cells ($n = 62, 58$) at initial biopsies, and patients without any events to indicate biopsy ($n = 112, 102$) are shown. (A to E) Mann-Whitney U tests and one-way ANOVA followed by Tukey's posttest, Jonckheere-Terpstra test, and Wilcoxon matched-pairs signed-rank test were used to compare the data. (F) Log-rank test was used to compare survival curves. (G) Gray's test was used to compare the cumulative incidence of nonrelapse mortality. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.005$. N.S., not significant.

Table 2. Characteristics of patients with goblet cell loss.

Variable	Severe goblet loss population	Mild or no goblet loss population	P
	n = 8	n = 65	
Age (median), years			0.81
	18–63 (49)	18–67 (47)	
Sex, n (%)			0.272
Male	3 (38)	39 (60)	
Female	5 (63)	26 (40)	
Underlying disease, n (%)			0.59
AML	1 (13)	17 (26)	
ALL	1 (13)	12 (18)	
ML	4 (50)	19 (29)	
MDS	1 (13)	9 (14)	
CML	0 (0)	2 (3)	
AA	0 (0)	4 (6)	
MM	0 (0)	1 (2)	
Others	1 (13)	1 (2)	
Disease risk index, n (%)			0.2430
Low	0 (0)	12 (18)	
Intermediate	6 (75)	27 (42)	
High	1 (13)	20 (31)	
Very high	1 (13)	6 (9)	
Donor source, n (%)			0.5000
BMT	4 (50)	40 (62)	
PBSCT	1 (13)	14 (22)	
CBT	3 (38)	11 (17)	
HLA match, n (%)			0.2640
Yes	2 (25)	33 (51)	
No	6 (75)	32 (49)	
Conditioning, n (%)			0.4600
MAC	5 (63)	29 (45)	
RIC	3 (38)	36 (55)	
TBI, n (%)			0.0569
Yes	6 (75)	63 (97)	
No	2 (25)	2 (3)	
GVHD grade, n (%)			0.0078
0–2	2 (25)	49 (75)	
3–4	6 (75)	16 (25)	
GI-GVHD stage, n (%)			0.0571
0	0 (0)	23 (35)	
1–2	5 (63)	32 (49)	
3–4	3 (38)	10 (15)	

$P = 0.00236$, respectively; Fig. 5G). Together, we concluded that goblet cells in the large intestine were specifically damaged in GI-GVHD, and severe goblet cell loss was associated with poor posttransplant outcomes.

DISCUSSION

Emerging evidence suggests that gut microbiota influences outcomes of allogeneic HSCT, namely, GVHD and graft versus leukemia (GVL) (4–6, 24). Especially, expansion of pathogenic microbes and reduction

of butyrate-producing commensal anaerobes that could directly protect the gut epithelial cells or induce regulatory T cells are associated with poor posttransplant prognosis (5, 22, 25, 26). Mucosal membranes of the small and large intestines protect hosts against bacterial translocation. Mucus layers of mucosal membrane cover the surface of mucosal membrane of GI tract and act as physical and chemical barriers against bacteria, suggesting that the integrity of mucus layers is a profound factor to determine the transplant outcomes after allo-HSCT, which induces gut dysbiosis (27). AMPs play a critical role to maintain mucosal homeostasis in the intestine by maintaining healthy intestinal microbiota or by direct protective activities on mucosal epithelium (6, 10, 28). α -Defensins are produced from Paneth cells in the small intestines and exert selective bactericidal effects against pathological bacteria while sparing beneficial commensal bacteria (9). We previously reported that GVHD targets Paneth cells and reduces production of α -defensins, leading to expansion of pathological bacteria and subsequent aggravation of GVHD (6). In the current study, we found that goblet cells were persistently reduced in GVHD, leading to the disruption of the mucus layer and bacterial translocation.

After production by epithelial cells, antimicrobial molecules diffuse into the mucus layer and are then released to the intestinal lumen. Thus, the mucus layer contains high concentrations of antimicrobial molecules and acts as chemical barrier against bacteria residing in the intestinal lumen. Some antimicrobial molecules were shown to act mainly in the mucus layer. Vaishnava *et al.* (13) reported that REGIII γ is produced in response to bacterial stimulation in the intestinal epithelial cells and plays a critical role to eradicate bacteria in the mucus layer. Okumura *et al.* (14) reported that Lypd8 is produced in the enterocytes in the colon and prevents bacterial translocation by inhibiting bacterial motility in the mucus layer. These findings suggest that mucus produced by goblet cells may maximize the antibacterial effects of a broader range of antimicrobial molecules, and disruption of this barrier function in GVHD resulted in increased bacterial translocation. In humans, LYPD8 inhibits motility of flagellated bacteria and its expression is reduced in the colonic mucosa of patients with ulcerative colitis (14, 29). However, the role of LYPD8 in clinical GVHD remains to be investigated. Expansion of a mucus-degrading bacterium, *Akkermansia muciniphila*, was associated with exaggeration of experimental GVHD, further emphasizing the critical role of the mucus layers in pathophysiology of GI-GVHD (30). We have recently reported that GVHD induced Paneth cell injury and subsequent reduction of Paneth cell-derived AMPs, leading to intestinal dysbiosis characterized by a reduction of microbial diversity without increase of bacterial load (6, 10). Given the major influence of Paneth cells on intestinal microbial diversity and because Paneth cells were markedly diminished in our models, we were not able to investigate a direct correlation between goblet cell loss and loss of microbial diversity. We have previously shown that administration of the epithelial growth factor R-Spondin1 protected Paneth cells and their production of α -defensins against experimental GVHD ameliorating intestinal dysbiosis and GVHD, whereas oral administration of mouse α -defensin cryptdin-4 only transiently ameliorated intestinal dysbiosis and GVHD (2, 10). R-Spondin1 expanded not only Paneth cells but also goblet cells, suggesting that administration of R-Spondin1, not like administration of AMPs, might prevent disruption of the mucus layer, which could be required for maximizing bactericidal effects of α -defensin.

We found that pre-SCT administration of IL-25 increased mature intestinal goblet cells that persisted after SCT, leading to reduced bacterial translocation and production of proinflammatory cytokines and ameliorated GVHD. We also found that the protective role of IL-25 was dependent on host production of Lypd8. Because Lypd8 is produced by intestinal epithelial cells other than goblet cells, IL-25 did not affect Lypd8 expression directly (14). Instead, IL-25 protects goblet cells and preserves the inner mucus layer, which plays a critical role to form the Lypd8-rich layer and inhibit bacterial translocation. IL-25 is produced by Tuft cells in the intestine and stimulates ILC2 to produce goblet cell growth factors (20). ILCs play a protective role against GVHD. Hanash *et al.* (31) showed that GVHD targets radioresistant ILC3 and the loss of ILC3 was associated with decreased production of epithelial growth factor IL-22 and disruption of the mucosal barrier. Unlike ILC3, ILC2s are profoundly reduced for prolonged period after conditioning regardless of GVHD development. Adoptive transfer of donor ILC2 harvested from IL-25-treated mice ameliorates GVHD by inducing myeloid-derived suppressor cells in the gut (23). Although we did not evaluate the fate of ILC2s after HSCT, we suspect that expanded ILC2s after IL-25 administration were depleted by irradiation leaving expanded radio-resistant goblet cells after HSCT. In humans, an activated phenotype of ILC2s and ILC3s in the peripheral blood is associated with reduced susceptibility to GVHD (32). It remains to be elucidated if an activated ILC2 phenotype correlates with goblet cell expansion and maturation in these patients.

Administration of epithelial growth factors could be a promising strategy to protect against GVHD. Previous reports demonstrated that IL-22 produced by ILC3 enhances epithelial regeneration from intestinal stem cells and plays a protective role against GVHD (31, 33, 34). Administration of IL-22 even after allogeneic HSCT is protective against GVHD (34). The antiviral cytokine type I IFN is another epithelial growth factor that enhances epithelial turnover and repair (35). RIG-I/MAVS and STING signaling stimulates production of type I IFN and strengthens gut integrity during irradiation- and GVHD-mediated tissue injury via IFN-I receptor signaling in intestinal epithelial cells (36). Exogenous type I IFN administrated on day -1 of HSCT also mitigates epithelial damages after HSCT (37). Indoles derived from intestinal microbiota up-regulate genes associated with the type I IFN response, which has been shown to protect against radiation-induced intestinal damage and reduce subsequent GVHD (38). In the current study, we showed that IL-25 administrated before HSCT was protective against GVHD. However, IL-25 administrated early after HSCT did not affect GVHD. This is probably due to profound depletion of ILC2 by pretransplant conditioning regimens because IL-25 requires ILC2 to expand goblet cells. Thus, strategies that enable to expand goblet cells independently of ILC2 are required to treat established GVHD. In addition, it remains to be explored if IL-25 effects could not be diminished by the post-transplant use of calcineurin inhibitors.

We found that goblet cells were specifically reduced in human GVHD but not in other diarrheal diseases developed after HSCT, and that degree of goblet cell loss was associated with severity of GVHD. Recently, it has been reported that Paneth cell count in duodenal biopsy correlates with severity of GI-GVHD (11). The frequency of goblet cells could be another simple diagnostic and monitoring biomarker of intestinal GVHD. We found that goblet cells persisted in patients with idiopathic diarrhea or CMV colitis. Further studies are required to assess fate of goblet cells in other conditions

that need to be differentiated from GVHD such as thrombotic microangiopathy and Epstein-Barr virus (EBV)-associated colitis. The recent development of confocal laser endomicroscopy made it possible to enumerate goblet cells without histological examination (39, 40). Endoscopic and histological evaluation of goblet cells may serve as a quick and simple diagnostic, monitoring, and prognostic indicator of intestinal GVHD, and goblet cell-protective strategies may benefit transplantation recipients by controlling microbiota via formation of antimicrobial molecules containing mucus layers and segregating bacteria from mucosal membrane.

MATERIALS AND METHODS

Study design

This study was initiated to explore the pathophysiological role of intestinal goblet cells in GVHD. To achieve this aim, we used both experimental mouse models of GVHD and human colon biopsy samples. For animal experiments, 8- to 12-week-old female mice were allocated randomly for each experimental group, ensuring that the mean body weight in each group was similar. Experiments were performed in a nonblinded fashion. All animal experiments were performed under the auspices of the Institutional Animal Care and Research Advisory Committee (approval number 12-0106). Histopathological analyses, qPCR, and measurement of proinflammatory cytokines were performed to evaluate goblet cell damage, bacterial translocation, Lypd8 expression, and severity of inflammation. GVHD severity was assessed by survival curves, clinical GVHD scores, and body weight change after HSCT. All experiments were performed at least two times. For analyses of human samples, we retrospectively reviewed medical records of 205 consecutive patients who underwent their first allogeneic HSCT at Hokkaido University Hospital between 2009 and 2015. We evaluated goblet cell damage in hematoxylin and eosin (H&E)-stained sections from 88 patients who underwent colon biopsy before or after HSCT. This study was approved by Institutional Review Board of Hokkaido University Hospital (approval number 016-0380) and conducted in accordance with the Declaration of Helsinki. Primary data are reported in data file S1.

Statistical analysis

Data were shown as means $\pm$ SE. Data were checked for normal distribution and similar variance between groups. Mann-Whitney *U* tests or one-way analysis of variance (ANOVA) followed by Tukey's posttest was used to compare data between groups. Jonckheere-Terpstra test was used to evaluate correlation between severity of GI-GVHD and degree of goblet cell loss. Wilcoxon matched-pairs signed-rank test was used to compare between initial and follow-up goblet cell numbers. To evaluate the impact of goblet cell loss in the large intestine after allogeneic transplantation on clinical outcomes, we used landmark method to exclude bias that may have arisen from including patients who died or had a relapse too early to develop severe goblet loss in the group without severe goblet loss (41, 42). To evaluate the impact of goblet cell loss on overall survival, we analyzed the patients who achieved engraftment and were surviving at 100 days after transplantation as a landmark point. The Kaplan-Meier product limit method was used to obtain survival probability, and the log-rank test was applied to compare survival curves. Nonrelapse mortality was defined as death without relapse. To evaluate the impact of goblet cell loss on the incidence of nonrelapse mortality, we ana-

lyzed the patients who achieved engraftment and were surviving without relapse at 100 days after transplantation as a landmark point. Incidence estimates for nonrelapse mortality were evaluated by using cumulative incidence method considering each other event as a competing risk event. The Gray test was used to compare the arms. Analyses were performed using Prism version 7.0 (GraphPad Software) and EZR version 3.4.1 (Saitama Medical Center, Jichi Medical University) (43). *P* < 0.05 was considered statistically significant.

SUPPLEMENTARY MATERIALS

stm.sciencemag.org/cgi/content/full/12/550/eaaw0720/DC1

Materials and Methods

Fig. S1. Plasma concentrations of IL-25 are elevated after HSCT.

Fig. S2. Pre-HSCT administration of IL-25 ameliorates GVHD in a Lypd8-dependent manner.

Fig. S3. Classification of biopsy specimens.

Fig. S4. Goblet cells in the colon are reduced in pathologically confirmed GI-GVHD.

Data file S1. Primary data.

References (44–51)

[View/request a protocol for this paper from Bio-protocol.](https://www.science.org at University of Utah on May 07, 2026)

REFERENCES AND NOTES

1. J. L. Ferrara, J. E. Levine, P. Reddy, E. Holler, Graft-versus-host disease. *Lancet* **373**, 1550–1561 (2009).
2. S. Takashima, M. Kadowaki, K. Aoyama, M. Koyama, T. Oshima, K. Tomizuka, K. Akashi, T. Teshima, The Wnt agonist R-spondin1 regulates systemic graft-versus-host disease by protecting intestinal stem cells. *J. Exp. Med.* **208**, 285–294 (2011).
3. J. Qin, R. Li, J. Raes, M. Arumugam, K. S. Burgdorf, C. Manichanh, T. Nielsen, N. Pons, F. Levenez, T. Yamada, D. R. Mende, J. Li, J. Xu, S. Li, D. Li, J. Cao, B. Wang, H. Liang, H. Zheng, Y. Xie, J. Tap, P. Lepage, M. Bertalan, J.-M. Batto, T. Hansen, D. L. Paslier, A. Linneberg, H. B. Nielsen, E. Pelletier, P. Renault, T. Sicheritz-Ponten, K. Turner, H. Zhu, C. Yu, S. Li, M. Jian, Y. Zhou, Y. Li, X. Zhang, S. Li, N. Qin, H. Yang, J. Wang, S. Brunak, J. Doré, F. Guarner, K. Kristiansen, O. Pedersen, J. Parkhill, J. Weissenbach; MetaHIT Consortium, P. Bork, S. D. Ehrlich, J. Wang, A human gut microbial gene catalogue established by metagenomic sequencing. *Nature* **464**, 59–65 (2010).
4. R. R. Jenq, C. Ubeda, Y. Taur, C. C. Menezes, R. Khanin, J. A. Dudakov, C. Liu, M. L. West, N. V. Singer, M. J. Equinda, A. Gobourne, L. Lipuma, L. F. Young, O. M. Smith, A. Ghosh, A. M. Hanash, J. D. Goldberg, K. Aoyama, B. R. Blazar, E. G. Pamer, M. R. M. van den Brink, Regulation of intestinal inflammation by microbiota following allogeneic bone marrow transplantation. *J. Exp. Med.* **209**, 903–911 (2012).
5. R. R. Jenq, Y. Taur, S. M. Devlin, D. M. Ponce, J. D. Goldberg, K. F. Ahr, E. R. Littmann, L. Ling, A. C. Gobourne, L. C. Miller, M. D. Docampo, J. U. Peled, N. Arpaia, J. R. Cross, T. K. Peets, M. A. Lumish, Y. Shono, J. A. Dudakov, H. Poeck, A. M. Hanash, J. N. Barker, M.-A. Perales, S. A. Giral, E. G. Pamer, M. R. M. van den Brink, Intestinal *blautia* is associated with reduced death from graft-versus-host disease. *Biol. Blood Marrow Transplant.* **21**, 1373–1383 (2015).
6. Y. Eriguchi, S. Takashima, H. Oka, S. Shimoji, K. Nakamura, H. Uryu, S. Shimoda, H. Iwasaki, N. Shimono, T. Ayabe, K. Akashi, T. Teshima, Graft-versus-host disease disrupts intestinal microbial ecology by inhibiting Paneth cell production of α -defensins. *Blood* **120**, 223–231 (2012).
7. N. H. Salzman, K. Hung, D. Haribhai, H. Chu, J. Karlsson-Sjöberg, E. Amir, P. Tegatz, M. Barman, M. Hayward, D. Eastwood, M. Stoel, Y. Zhou, E. Sodergren, G. M. Weinstock, C. L. Bevins, C. B. Williams, N. A. Bos, Enteric defensins are essential regulators of intestinal microbial ecology. *Nat. Immunol.* **11**, 76–83 (2010).
8. A. J. Ouellette, M. M. Hsieh, M. T. Nosek, D. F. Cano-Gauci, K. M. Huttner, R. N. Buick, M. E. Selsted, Mouse Paneth cell defensins: Primary structures and antibacterial activities of numerous cryptdin isoforms. *Infect. Immun.* **62**, 5040–5047 (1994).
9. K. Masuda, N. Sakai, K. Nakamura, S. Yoshioka, T. Ayabe, Bactericidal activity of mouse α -defensin cryptdin-4 predominantly affects noncommensal bacteria. *J. Innate Immun.* **3**, 315–326 (2011).
10. E. Hayase, D. Hashimoto, K. Nakamura, C. Noizat, R. Ogasawara, S. Takahashi, H. Ohigashi, Y. Yokoi, R. Sugimoto, S. Matsuoka, T. Ara, E. Yokoyama, T. Yamakawa, K. Ebata, T. Kondo, R. Hiramine, T. Aizawa, Y. Ogura, T. Hayashi, H. Mori, K. Kurokawa, K. Tomizuka, T. Ayabe, T. Teshima, R-spondin1 expands Paneth cells and prevents dysbiosis induced by graft-versus-host disease. *J. Exp. Med.* **214**, 3507–3518 (2017).
11. J. E. Levine, E. Huber, S. T. G. Hammer, A. C. Harris, J. K. Greenson, T. M. Braun, J. L. M. Ferrara, E. Holler, Low Paneth cell numbers at onset of gastrointestinal GVHD identify patients at high risk for non-relapse mortality. *Blood* **122**, 1505–1509 (2013).
12. R. Sender, S. Fuchs, R. Milo, Revised estimates for the number of human and bacteria cells in the body. *PLOS Biol.* **14**, e1002533 (2016).

13. S. Vaishnava, M. Yamamoto, K. M. Severson, K. A. Ruhn, X. Yu, O. Koren, R. Ley, E. K. Wakeland, L. V. Hooper, The antibacterial lectin RegIII γ promotes the spatial segregation of microbiota and host in the intestine. *Science* **334**, 255–258 (2011).
14. R. Okumura, T. Kurakawa, T. Nakano, H. Kayama, M. Kinoshita, D. Motooka, K. Gotoh, T. Kimura, N. Kamiyama, T. Kusu, Y. Ueda, H. Wu, H. Iijima, S. Barman, H. Osawa, H. Matsuno, J. Nishimura, Y. Ohba, S. Nakamura, T. Iida, M. Yamamoto, E. Umemoto, K. Sano, K. Takeda, Lypd8 promotes the segregation of flagellated microbiota and colonic epithelia. *Nature* **532**, 117–121 (2016).
15. L. Antoni, S. Nuding, D. Weller, M. Gersemann, G. Ott, J. Wehkamp, E. F. Stange, Human colonic mucus is a reservoir for antimicrobial peptides. *J. Crohns Colitis* **7**, e652–e664 (2013).
16. M. L. Eigenbrodt, E. H. Eigenbrodt, D. L. Thiele, Histologic similarity of murine colonic graft-versus-host disease (GVHD) to human colonic GVHD and inflammatory bowel disease. *Am. J. Pathol.* **137**, 1065–1076 (1990).
17. D. A. Levy, A. Wefald, Gut mucosal mast cells and goblet cells during acute graft-versus-host disease in rats. *Ann. Inst. Pasteur Immunol.* **137D**, 281–288 (1986).
18. R. Nowarski, R. Jackson, N. Gagliani, M. R. de Zoete, N. W. Palm, W. Bailis, J. S. Low, C. C. D. Harman, M. Graham, E. Elinav, R. A. Flavell, Epithelial IL-18 equilibrium controls barrier function in colitis. *Cell* **163**, 1444–1456 (2015).
19. S. A. Saenz, M. C. Siracusa, L. A. Monticelli, C. G. K. Ziegler, B. S. Kim, J. R. Brestoff, L. W. Peterson, E. J. Wherry, A. W. Goldrath, A. Bhandoola, D. Artis, IL-25 simultaneously elicits distinct populations of innate lymphoid cells and multipotent progenitor type 2 (MPP^{type2}) cells. *J. Exp. Med.* **210**, 1823–1837 (2013).
20. J. von Moltke, M. Ji, H.-E. Liang, R. M. Locksley, Tuft-cell-derived IL-25 regulates an intestinal ILC2-epithelial response circuit. *Nature* **529**, 221–225 (2016).
21. M. M. Heimesaat, A. Nogai, S. Bereswill, R. Plickert, A. Fischer, C. Loddikenkemper, U. Steinhoff, S. Tchapchet, E. Thiel, M. A. Freudenberg, U. B. Göbel, L. Uharek, MyD88/TLR9 mediated immunopathology and gut microbiota dynamics in a novel murine model of intestinal graft-versus-host disease. *Gut* **59**, 1079–1087 (2010).
22. Y. Taur, R. R. Jenq, M.-A. Perales, E. R. Littmann, S. Morjaria, L. Ling, D. No, A. Gobourne, A. Viale, P. B. Dahi, D. M. Ponce, J. N. Barker, S. Giral, M. van den Brink, E. G. Pamer, The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation. *Blood* **124**, 1174–1182 (2014).
23. D. W. Bruce, H. E. Stefanski, B. G. Vincent, T. A. Dant, S. Reisdorf, H. Bommasamy, D. A. Serody, J. E. Wilson, K. P. McKinnon, W. D. Shlomchik, P. M. Armistead, J. P. Y. Ting, J. T. Woosley, B. R. Blazar, D. M. W. Zaiss, A. N. J. McKenzie, J. M. Coghill, J. S. Serody, Type 2 innate lymphoid cells treat and prevent acute gastrointestinal graft-versus-host disease. *J. Clin. Invest.* **127**, 1813–1825 (2017).
24. J. U. Peled, S. M. Devlin, A. Staffas, M. Lumish, R. Khanin, E. R. Littmann, L. Ling, S. Kosuri, M. Maloy, J. B. Slingerland, K. F. Ahr, K. A. Porosnicu Rodriguez, Y. Shono, A. E. Slingerland, M. D. Docampo, A. D. Sung, D. Weber, A. M. Alousi, B. Gyurkocza, D. M. Ponce, J. N. Barker, M.-A. Perales, S. A. Giral, Y. Taur, E. G. Pamer, R. R. Jenq, M. R. M. van den Brink, Intestinal microbiota and relapse after hematopoietic-cell transplantation. *J. Clin. Oncol.* **35**, 1650–1659 (2017).
25. K. Atarashi, T. Tanoue, K. Oshima, W. Suda, Y. Nagano, H. Nishikawa, S. Fukuda, T. Saito, S. Narushima, K. Hase, S. Kim, J. V. Fritz, P. Wilmes, S. Ueha, K. Matsushima, H. Ohno, B. Olle, S. Sakaguchi, T. Taniguchi, H. Morita, M. Hattori, K. Honda, T_{reg} induction by a rationally selected mixture of Clostridia strains from the human microbiota. *Nature* **500**, 232–236 (2013).
26. N. D. Mathewson, R. Jenq, A. V. Mathew, M. Koenigsnecht, A. Hanash, T. Toubai, K. Oravec-Wilson, S.-R. Wu, Y. Sun, C. Rossi, H. Fujiwara, J. Byun, Y. Shono, C. Lindemans, M. Calafiore, T. M. Schmidt, K. Honda, V. B. Young, S. Pennathur, M. van den Brink, P. Reddy, Gut microbiome-derived metabolites modulate intestinal epithelial cell damage and mitigate graft-versus-host disease. *Nat. Immunol.* **17**, 505–513 (2016).
27. J. U. Peled, A. L. C. Gomes, S. M. Devlin, E. R. Littmann, Y. Taur, A. D. Sung, D. Weber, D. Hashimoto, A. E. Slingerland, J. B. Slingerland, M. Maloy, A. G. Clurman, C. K. Stein-Thoeringer, K. A. Markey, M. D. Docampo, M. B. da Silva, N. Khan, A. Gessner, J. A. Messina, K. Romero, M. V. Lew, A. Bush, L. Bohannon, D. G. Brereton, E. Fontana, L. A. Amoretti, R. J. Wright, G. K. Armijo, Y. Shono, M. Sanchez-Escamilla, N. C. Flores, A. A. Tomas, R. J. Lin, L. Y. S. Segundo, G. L. Shah, C. Cho, M. Scordo, I. Politikos, K. Hayasaka, Y. Hasegawa, B. Gyurkocza, D. M. Ponce, J. N. Barker, M. A. Perales, S. A. Giral, R. R. Jenq, T. Teshima, N. J. Chao, E. Holler, J. B. Xavier, E. G. Pamer, M. R. M. van den Brink, Microbiota as predictor of mortality in allogeneic hematopoietic-cell transplantation. *N. Engl. J. Med.* **382**, 822–834 (2020).
28. D. Zhao, Y.-H. Kim, S. Jeong, J. K. Greenson, M. S. Chaudhry, M. Hoepting, E. R. Anderson, M. R. van den Brink, J. U. Peled, A. L. Gomes, A. E. Slingerland, M. J. Donovan, A. C. Harris, J. E. Levine, U. Ozbek, L. V. Hooper, T. S. Stappenbeck, A. Ver Heul, T.-C. Liu, P. Reddy, J. L. Ferrara, Survival signal REG3 α prevents crypt apoptosis to control acute gastrointestinal graft-versus-host disease. *J. Clin. Invest.* **128**, 4970–4979 (2018).
29. C.-C. Hsu, R. Okumura, K. Takeda, Human LYPD8 protein inhibits motility of flagellated bacteria. *Inflamm. Regen.* **37**, 23 (2017).
30. Y. Shono, M. D. Docampo, J. U. Peled, S. M. Perobelli, E. Velardi, J. J. Tsai, A. E. Slingerland, O. M. Smith, L. F. Young, J. Gupta, S. R. Lieberman, H. V. Jay, K. F. Ahr, K. A. P. Rodriguez, K. Xu, M. Calafiore, H. Poeck, S. Caballero, S. M. Devlin, F. Rapaport, J. A. Dudakov, A. M. Hanash, B. Gyurkocza, G. F. Murphy, C. Gomes, C. Liu, E. L. Moss, S. B. Falconer, A. S. Bhatt, Y. Taur, E. G. Pamer, M. R. M. van den Brink, R. R. Jenq, Increased GVHD-related mortality with broad-spectrum antibiotic use after allogeneic hematopoietic stem cell transplantation in human patients and mice. *Sci. Transl. Med.* **8**, 339ra371 (2016).
31. A. M. Hanash, J. A. Dudakov, G. Hua, M. H. O'Connor, L. F. Young, N. V. Singer, M. L. West, R. R. Jenq, A. M. Holland, L. W. Kappel, A. Ghosh, J. J. Tsai, U. K. Rao, N. L. Yim, O. M. Smith, E. Velardi, E. B. Hawryluk, G. F. Murphy, C. Liu, L. A. Fouser, R. Kolesnick, B. R. Blazar, M. R. M. van den Brink, Interleukin-22 protects intestinal stem cells from immune-mediated tissue damage and regulates sensitivity to graft versus host disease. *Immunity* **37**, 339–350 (2012).
32. J. M. Munneke, A. T. Björklund, J. M. Mjösberg, K. Garming-Legert, J. H. Bernink, B. Blom, C. Huisman, M. H. J. van Oers, H. Spits, K.-J. Malmberg, M. D. Hazenberg, Activated innate lymphoid cells are associated with a reduced susceptibility to graft-versus-host disease. *Blood* **124**, 812–821 (2014).
33. M. Couturier, B. Lamarthée, J. Arbez, J.-C. Renauld, C. Bossard, F. Malard, F. Bonnefoy, M. Mohy, S. Perruche, P. Tiberghien, P. Saas, B. Gaugler, IL-22 deficiency in donor T cells attenuates murine acute graft-versus-host disease mortality while sparing the graft-versus-leukemia effect. *Leukemia* **27**, 1527–1537 (2013).
34. C. A. Lindemans, M. Calafiore, A. M. Mertelsmann, M. H. O'Connor, J. A. Dudakov, R. R. Jenq, E. Velardi, L. F. Young, O. M. Smith, G. Lawrence, J. A. Ivanov, Y.-Y. Fu, S. Takashima, G. Hua, M. L. Martin, K. P. O'Rourke, Y.-H. Lo, M. Mokry, M. Romera-Hernandez, T. Cupedo, L. E. Dow, E. E. Nieuwenhuis, N. F. Shroyer, C. Liu, R. Kolesnick, M. R. M. van den Brink, A. M. Hanash, Interleukin-22 promotes intestinal-stem-cell-mediated epithelial regeneration. *Nature* **528**, 560–564 (2015).
35. L. Sun, H. Miyoshi, S. Origanti, T. J. Nice, A. C. Barger, N. A. Manieri, L. A. Fogel, A. R. French, D. Piwnica-Worms, H. Piwnica-Worms, H. W. Virgin, D. J. Lenschow, T. S. Stappenbeck, Type I interferons link viral infection to enhanced epithelial turnover and repair. *Cell Host Microbe* **17**, 85–97 (2015).
36. J. C. Fischer, M. Bscheider, G. Eisenkolb, C.-C. Lin, A. Wintges, V. Otten, C. A. Lindemans, S. Heidegger, M. Rudelius, S. Monette, K. A. P. Rodriguez, M. Calafiore, S. Lieberman, C. Liu, S. Lienenklaus, S. Weiss, U. Kalinke, J. Ruland, C. Peschel, Y. Shono, M. Docampo, E. Velardi, R. R. Jenq, A. M. Hanash, J. A. Dudakov, T. Haas, M. R. M. van den Brink, H. Poeck, RIG-I/MAVS and STING signaling promote gut integrity during irradiation- and immune-mediated tissue injury. *Sci. Transl. Med.* **9**, eaag2513 (2017).
37. R. J. Robb, E. Kreijveld, R. D. Kuns, Y. A. Wilson, S. D. Olver, A. L. J. Don, N. C. Raffelt, N. A. De Weerd, K. E. Lineburg, A. Varelias, K. A. Markey, M. Koyama, A. D. Clouston, P. J. Hertzog, K. P. A. Macdonald, G. R. Hill, Type I IFNs control GVHD and GVL responses after transplantation. *Blood* **118**, 3399–3409 (2011).
38. A. Swimm, C. R. Giver, Z. DeFilipp, S. Rangaraju, A. Sharma, A. Ulezko Antonova, R. Sonowal, C. Capaldo, D. Powell, M. Qayed, D. Kaiman, E. K. Waller, Indoles derived from intestinal microbiota act via type I interferon signaling to limit graft-versus-host disease. *Blood* **132**, 2506–2519 (2018).
39. A. V. Arya, B. M. Yan, Ultra high magnification endoscopy: Is seeing really believing? *World J. Gastrointest. Endosc.* **4**, 462–471 (2012).
40. D. Boschetto, H. Mirzaei, R. W. L. Leong, E. Grisan, Detection and density estimation of goblet cells in confocal endoscopy for the evaluation of celiac disease. *Conf. Proc. IEEE Eng. Med. Biol. Soc.* **2015**, 6248–6251 (2015).
41. J. R. Anderson, K. C. Cain, R. D. Gelber, Analysis of survival by tumor response. *J. Clin. Oncol.* **1**, 710–719 (1983).
42. J. P. Klein, J. D. Rizzo, M. J. Zhang, N. Keiding, Statistical methods for the analysis and presentation of the results of bone marrow transplants. Part 2: Regression modeling. *Bone Marrow Transplant.* **28**, 1001–1011 (2001).
43. Y. Kanda, Investigation of the freely available easy-to-use software 'EZ' for medical statistics. *Bone Marrow Transplant.* **48**, 452–458 (2013).
44. H. Uryu, D. Hashimoto, K. Kato, E. Hayase, S. Matsuoka, R. Ogasawara, S. Takahashi, Y. Maeda, H. Iwasaki, T. Miyamoto, S. Saijo, Y. Iwakura, G. R. Hill, K. Akashi, T. Teshima, α -Mannan induces Th17-mediated pulmonary graft-versus-host disease in mice. *Blood* **125**, 3014–3023 (2015).
45. R. Ogasawara, D. Hashimoto, S. Kimura, E. Hayase, T. Ara, S. Takahashi, H. Ohgashi, K. Yoshioka, T. Tatenoe, E. Yokoyama, K. Ebata, T. Kondo, J. Sugita, M. Onozawa, T. Iwanaga, T. Teshima, Intestinal lymphatic endothelial cells produce R-Spondin3. *Sci. Rep.* **8**, 10719 (2018).
46. K. R. Cooke, L. Kobzik, T. R. Martin, J. Brewer, J. Delmonte Jr., J. M. Crawford, J. L. Ferrara, An experimental model of idiopathic pneumonia syndrome after bone marrow transplantation: I. The roles of minor H antigens and endotoxin. *Blood* **88**, 3230–3239 (1996).
47. K. A. Alexander, R. Flynn, K. E. Lineburg, R. D. Kuns, B. E. Teal, S. D. Olver, M. Lor, N. A. Raffelt, M. Koyama, L. Leveque, L. Le Texier, M. Melino, K. A. Markey, A. Varelias, C. Engwerda, J. S. Serody, B. Janela, F. Ginhoux, A. D. Clouston, B. R. Blazar, G. R. Hill,

- K. P. MacDonald, CSF-1-dependant donor-derived macrophages mediate chronic graft-versus-host disease. *J. Clin. Invest.* **124**, 4266–4280 (2014).
48. M. Bogunovic, F. Ginhoux, J. Helft, L. Shang, D. Hashimoto, M. Greter, K. Liu, C. Jakubczik, M. A. Ingersoll, M. Leboeuf, E. R. Stanley, M. Nussenzweig, S. A. Lira, G. J. Randolph, M. Merad, Origin of the lamina propria dendritic cell network. *Immunity* **31**, 513–525 (2009).
49. A. W. Goodyear, A. Kumar, S. Dow, E. P. Ryan, Optimization of murine small intestine leukocyte isolation for global immune phenotype analysis. *J. Immunol. Methods* **405**, 97–108 (2014).
50. D. Przepiorka, D. Weisdorf, P. Martin, H. G. Klingemann, P. Beatty, J. Hovs, E. D. Thomas, 1994 consensus conference on acute GVHD grading. *Bone Marrow Transplant.* **15**, 825–828 (1995).
51. A. Kreft, A. Mottok, I. Mesteri, D. M. Cardona, A. Janin, A. A. Kühl, M. Andrusis, A. Brunner, H. M. Shulman, G. Negri, A. Tzankov, E. Huber; Gastrointestinal Pathology Group of the German-Austrian-Swiss GvHD Consortium, Consensus diagnostic histopathological criteria for acute gastrointestinal graft versus host disease improve interobserver reproducibility. *Virchows Arch.* **467**, 255–263 (2015).

Acknowledgments: We thank C. Yokoyama and T. Sekiguchi for technical assistance.

Funding: This study was supported by JSPS KAKENHI (17H04206 and 17K19641 to T.T. and

17K09945 to D.H.), the Center of Innovation Program from JST (T.T.), and Mochida Memorial Foundation for Medical and Pharmaceutical Research (D.H.). **Author contributions:** D.H. and T.T. developed the conceptual framework of the study, designed the experiments, conducted experiments, analyzed data, and wrote the paper. T.A. conducted experiments, analyzed data, and wrote the paper. E.H., C.N., R.K., Y.H., K. Matsuda, S.O., K.E., R. Ogasawara, S.T., H.O., E.Y., J.S., M.O., and R. Okumura conducted experiments. K. Matsuo performed statistical analyses. Y.M. and K.T. supervised experiments. **Competing interests:** The authors declare no competing financial interests. **Data and materials availability:** All data associated with this study are present in the paper or Supplementary Materials.

Submitted 19 November 2018

Resubmitted 7 January 2020

Accepted 27 May 2020

Published 1 July 2020

10.1126/scitranslmed.aaw0720

Citation: T. Ara, D. Hashimoto, E. Hayase, C. Noizat, R. Kikuchi, Y. Hasegawa, K. Matsuda, S. Ono, Y. Matsuno, K. Ebata, R. Ogasawara, S. Takahashi, H. Ohigashi, E. Yokoyama, K. Matsuo, J. Sugita, M. Onozawa, R. Okumura, K. Takeda, T. Teshima, Intestinal goblet cells protect against GVHD after allogeneic stem cell transplantation via Lypd8. *Sci. Transl. Med.* **12**, eaaw0720 (2020).

Intestinal goblet cells protect against GVHD after allogeneic stem cell transplantation via Lypd8

Takahide Ara, Daigo Hashimoto, Eiko Hayase, Clara Noizat, Ryo Kikuchi, Yuta Hasegawa, Kana Matsuda, Shoko Ono, Yoshihiro Matsuno, Ko Ebata, Reiki Ogasawara, Shuichiro Takahashi, Hiroyuki Ohigashi, Emi Yokoyama, Keitaro Matsuo, Junichi Sugita, Masahiro Onozawa, Ryu Okumura, Kiyoshi Takeda, and Takanori Teshima

Sci. Transl. Med. **12** (550), eaaw0720. DOI: 10.1126/scitranslmed.aaw0720

Guarding the gut against GVHD

The gastrointestinal tract is one of the major target organs of graft-versus-host disease (GVHD) following allogeneic hematopoietic stem cell transplant. Goblet cells, mucus-secreting epithelial cells, are diminished in GVHD, but the pathophysiological implications were unknown. Ara *et al.* used mouse GVHD models to demonstrate that goblet cell loss disrupted the inner mucus layer of the colon and permitted bacterial translocation. GVHD was also exacerbated in mice lacking the antimicrobial Lypd8. Goblet cell preservation with IL-25 pretreatment maintained the gut barrier and blunted GVHD. In patient biopsies, goblet cell loss was associated with severe gastrointestinal GVHD and poor outcomes. Collectively, these results suggest that retaining goblet cells with IL-25 conditioning could prevent GVHD in the gut.

View the article online

<https://www.science.org/doi/10.1126/scitranslmed.aaw0720>

Permissions

<https://www.science.org/help/reprints-and-permissions>

Use of this article is subject to the [Terms of service](#)

Science Translational Medicine (ISSN 1946-6242) is published by the American Association for the Advancement of Science. 1200 New York Avenue NW, Washington, DC 20005. The title *Science Translational Medicine* is a registered trademark of AAAS.

Copyright © 2020 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works